TEACHER GUIDE · KINDERGARTEN · UNPLUGGED THINKING

Break It Into Parts

Quick facilitation notes & answer key

Goal

Children see that a whole picture is built from smaller shape parts, and name/draw those parts. Breaking a whole into parts is decomposition — the computational-thinking skill of splitting a big problem into smaller, manageable pieces.

Kindergarten frame (Ontario)

Problem Solving and Innovating — children analyze how a whole is composed of parts.

No reading required of the child: read each prompt aloud; the child draws the parts.

Materials

The printed worksheet + a pencil/crayon. Optional: pattern blocks to build a house/rocket and then take it apart into its shapes.

Run it (about 15 min)

1. MODEL: show the rocket — 'a triangle on top, a square for the body. That is 2 parts.'
2. GUIDED: do the house in Exercise 1 together — square walls + triangle roof.
3. YOU TRY: children draw the snowman's 3 circles (Ex 2), then Bit the robot's 4 parts (Ex 3).
4. Connect: 'Coders break a big job into small steps — just like we broke the robot into parts.'

Answer key (modelled by the run-gate)

Model — rocket = 2 parts (1 triangle + 1 square).

Ex 1 — house = 2 parts (1 square + 1 triangle).

Ex 2 — snowman = 3 parts (3 circles).

Ex 3 — Bit the robot = 4 parts (3 squares + 1 circle).

Watch for & differentiate

Common slip: drawing the whole picture instead of separate parts — point to each empty box as one part.

Simplify: build the object with blocks, then lift each shape out and name it.

Extend: ask the child what other things are made of parts (a face, a flower, a car).

Movement option: build a 'robot' with bodies — children each become one part (head, body, feet).

Success looks like

The child draws the correct number of separate shape parts for each whole (2, 3, then 4).